

ALEX ARMENGOL URPI

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EDUCATION

Massachusetts Institute of Technology

PhD in Human-Computer Interfaces

- “La Caixa” Fellowship for studies at MIT (full tuition and stipend)

Cambridge, MA

2018 – June 2023

Massachusetts Institute of Technology

M.S., Mechanical Engineering.

- Concentration on Human-Computer Interfaces for IoT and AR/VR

Cambridge, MA

2016 - 2018

CFIS - Universitat Politècnica de Catalunya

Dual Major Bachelor's Degree:

- Telecommunications Engineering.
- Industrial Engineering.

- CFIS scholarship fellow to pursue 2 major engineering degrees simultaneously. 40 admitted students/year in Spain.

Barcelona, Spain

2011 - 2016

RESEARCH EXPERIENCE

MIT, Department of Mechanical Engineering

Research Scientist

Cambridge, MA

October 2025 - Present

- Developed a transformer-based open-vocabulary 3D grounding system for real-time object localization in AR environments. Fused point-cloud geometry with DINOv2 visual embeddings in a PyTorch architecture to enable zero-shot localization of unseen objects. Leveraged SLAM camera poses to improve accuracy and runtime efficiency in cluttered scenes. Comparable results to closed-set, fully supervised models.
- Built a real-time AR guidance system that uses LLMs to generate free-form 3D overlays from geometric primitives. Integrated VLMs (CLIP-based models), depth camera perception and AI-based object tracking to dynamically adapt LLM-generated overlays to object geometry, task context, and headset pose, enabling flexible spatial instructions beyond predefined AR guidance templates.

MIT, Department of Mechanical Engineering

Postdoctoral associate

Cambridge, MA

June 2023 – October 2025

- Developed a multimodal sensing framework to capture tacit knowledge from expert demonstrations in real-world tasks (crafts and sports). Instrumented experts, tools, and environment with sensors including eye tracking, EMG, accelerometers, airflow sensors, and scene cameras to capture behavioral and biomechanical signals. Built a data pipeline to generate instructional videos and AI-ready datasets for modeling differences between experts and novices.

MIT, Department of Mechanical Engineering

Research Assistant. Auto-ID Lab. Advisor: Sanjay E. Sarma

Cambridge, MA

2015 – June 2023

- Study of human implicit learning and tacit knowledge to enhance human expertise and AI performance:
 - Enhancing AI systems: Designed a conv-deconv neural net architecture to predict human gaze distributions, demonstrating that AI models benefit from tacit human knowledge signals beyond labeled data. Our system outperformed by 20+ points baseline image classification models when data is scarce.
 - Enhancing human performance: Identified gaze and attention patterns as markers of tacit expertise and generated expert attention heatmaps from eye-tracking data to guide learners, improving performance in visual tasks of trained individuals compared to an equally trained control group.
- Development of novel interfaces for foveal (overt) and peripheral (covert) visual attention tracking using EEG. Proposed and validated a first-of-its-kind SSVEP-based approach using single-frequency phase-coded stimuli to decode covert visual-spatial attention from EEG, achieving 88% accuracy across participants and enabling higher-capacity attention-tracking systems.
- Developed a hybrid EEG + eye-tracking system that leverages imperceptible high-frequency (≥ 40 Hz) visual stimuli and Bayesian fusion to improve gaze estimation accuracy, achieving +11% average gains and >45% improvement in peripheral regions over standard eye trackers.
- Designed and evaluated a novel SSVEP-based BCI that integrates high-frequency, imperceptible visual stimuli into VR environments to infer user attention from EEG, enabling seamless, hands-free navigation with real-time feedback.
- Developed a closed-loop, EEG-based system for objective assessment of visual function, using adaptive visual stimuli and deep learning to detect deficits in real time (e.g., glaucoma, optic nerve damage).

INDUSTRY EXPERIENCE

LIFEVERSE AI

Part time consultant

London, UK

Spring 2023

- Consulting on research directions on AI for behavioral understanding on children's attention for education

PHILIPS NORTH AMERICA

Augmented Reality Research Intern.

Cambridge, MA

Summer 2019

- Research on Augmented Reality-guided surgical procedures.

ALMATECH INC.

Co-leader of engineering department.

Barcelona, Spain

Oct 2014 – March 2015

- Worked on UPC-derived spinoff company devoted to the WPT domain.

SELECTED PUBLICATIONS

- [1] *HoLocate: An Open-Vocabulary, Zero-Shot 3D Grounding Framework for Augmented Reality*
Alex Armengol-Urpi, Dulanga Weerakoon, Andres Salazar, Sanjay Sarma
In preparation.
- [2] *Free Form 3D Overlays for Open-ended Augmented Reality Task Assistance Using LLMs*
Alex Armengol-Urpi, Dulanga Weerakoon, Andres Salazar, Sanjay Sarma
In preparation.
- [3] *Cognitive Reinforcement: Capturing Tacit Knowledge and Enhancing Expertise with a Biofeedback Interface for Visual Attention. Featured in MIT News.*
Alex Armengol-Urpi, Andres Salazar, Pawan Sinha, Sanjay Sarma
Journal of Neural Engineering. March 2026.
- [4] *Brain-Hack: Remotely Injecting False Brain-Waves with RF to Take Control of a Brain-Computer Interface*
Alex Armengol-Urpi, Reid Kovacs, Sanjay Sarma
Workshop on CPS&IoT Security and Privacy. **Best paper award.**
- [5] *Brainwave-Augmented Eye Tracker: High-Frequency SSVEPs Improves Camera-Based Eye Tracking Accuracy*
Alex Armengol-Urpi, Andres Salazar, Sanjay Sarma
IUI 2022.
- [6] *A system approach for closed-loop assessment of neuro-visual function based on convolutional neural network analysis of EEG signals*
Simon Stock, **Alex Armengol-Urpi**, Bálint Kovács, Heiko Maier, Marius Gerdes, Wilhelm Stork, Sanjay Sarma
SPIE Neurophotonics 2022
- [7] *Towards Industrial IoT-AR Systems Using Deep Learning-Based Object Pose Estimation*
Yongbin Sun, Sai Nithin Reddy Kantareddy, Joshua Siegel, **Alex Armengol-Urpi**, Xiaoyu Wu, Hongyu Wang, Sanjay Sarma
IPCCC 2019
- [8] *Magichand: Interact with IoT Devices in Augmented Reality Environment*
Yongbin Sun, **Alex Armengol-Urpi**, Sai Nithin Reddy Kantareddy, Joshua Siegel, Sanjay Sarma
IEEE VR 2019
- [9] *Sublime: a Hands-Free Virtual Reality Menu Navigation System Using a High-Frequency SSVEP-based Brain-Computer Interface*
Alex Armengol-Urpi, Sanjay Sarma
VRST 2018

Patents

- [10] Sanjay E Sarma, Rahul Bhattacharyya, Joshua E Siegel, **Alex Armengol-Urpi**, Pankhuri Sen, Sai Nithin Reddy Kantareddy. "Antenna and system for wireless sensing of health monitoring"
- [11] Sanjay E Sarma, **Alex Armengol-Urpi**. "Brain-computer interface for user's visual focus detection"

FELLOWSHIPS, AWARDS & HONORS

- **MIT News feature.** Research on tacit knowledge and visual attention highlighted by *MIT News*, 2026.
- **Best paper award.** Workshop on CPS&IoT Security and Privacy, 2023.
- **"La Caixa" Fellowship.** Full funding for 2 years of PhD program at MIT, 2018.
- **CFIS Fellow.** Telecom. Engineering Degree tuition covered by the Technical University of Catalonia, 2011.
- **Youth Research Awards.** Award given by the Government of Catalonia for my high school final thesis, 2010.
- **Gov. of Catalonia Fellow.** 1st year college tuition covered by Gov. of Catalonia for Honors in high school, 2010.
- **Catalunya Caixa Bank Fellow.** For outstanding score in the National University entry exam (13.48/14), 2010.